ContraLegal-AI: An Intelligent AI Paralegal for Automated Contract Risk Analysis using Transformer Models, Retrieval-Augmented Generation, and Spatial NLP

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***Abstract*—Manual review of commercial contracts is a resource-intensive, error-prone, and expensive process that limits access to accurate legal risk assessment. ContraLegal-AI presents a full-stack intelligent legal-tech platform that automates the extraction, classification, explanation, and visual annotation of risk within PDF contracts. The system's architecture comprises three synergistic components: (1) a fine-tuned `nlpaueb/legal-bert-base-uncased` transformer for three-class risk classification (Low, Medium, High Risk), achieving 97.01% accuracy and a macro ROC-AUC of 0.9948—surpassing a TF-IDF + Random Forest baseline by 2.57%; (2) a Retrieval-Augmented Generation (RAG) pipeline using FAISS vector storage and LangChain for clause-level explanation, rewriting, and conversational contract querying; and (3) a Spatial NLP module leveraging PyMuPDF's coordinate-mapping API to physically annotate risky clauses within the original PDF document. A comprehensive ablation study demonstrates that Legal-BERT provides a critical +11.98% improvement in High Risk Recall over classical methods—the most consequential metric in legal risk detection. The system is deployed as an interactive Streamlit application with full CI/CD via GitHub Actions.**

***Index Terms*—Legal NLP, Transformer Fine-tuning, Legal-BERT, RAG, FAISS, Risk Classification, Contract Analysis, Spatial NLP, PyMuPDF, LangChain.**

I. Introduction

In modern commercial environments, legal teams must process vast volumes of contracts—Non-Disclosure Agreements (NDAs), Master Services Agreements (MSAs), Employment Contracts, and Merger and Acquisition documents—each containing clauses that may impose significant financial liability, asymmetric obligation, or unfavorable jurisdiction. The manual identification of such high-risk clauses demands deep legal expertise and is prohibitively time-consuming.

Prior work in legal text classification has predominantly relied on binary labeling schemes and bag-of-words statistical methods [1]. While effective for shallow risk detection, these approaches fail to capture the semantic nuance that differentiates a "Medium Risk" indemnity clause from a "High Risk" unlimited liability clause. Furthermore, they provide no mechanism for explanation, rewriting, or interactive querying.

ContraLegal-AI addresses this gap by delivering an end-to-end automated paralegal system. The core contributions of this work are:

1) A fine-tuned Legal-BERT transformer delivering three-class risk classification with superior performance over classical ML baselines, validated through a formal ablation study.
2) A RAG-based conversational AI engine allowing users to interrogate contracts in natural language, explain flagged clauses, and generate fairer rewrites.
3) A Spatial NLP highlighting system that outputs a physically annotated PDF with risky clauses marked via coordinate-to-text mapping.
4) A production-grade deployment with automated CI/CD testing via GitHub Actions and a streamlit web interface.

II. Related Work

Early legal document automation relied on rule-based systems and regular expression matching for clause identification [1]. These lacked generalizability across diverse contract types. The introduction of neural embeddings and BERT-family models transformed the field [2]. `legal-bert-base-uncased`, pre-trained on 12 GB of diverse English legal text (legislation, court decisions, contracts), has demonstrated superior domain adaptation compared to general-purpose BERT variants [3].

For retrieval-augmented systems, Lewis et al. [4] demonstrated that grounding LLM responses in retrieved documents significantly reduces hallucination—a critical requirement in legal contexts. The Contract Understanding Atticus Dataset (CUAD) [5] has become the standard benchmark for supervised contract clause classification. Our approach distinguishes itself by combining a domain-specific transformer for classification with RAG for generation, together with a novel spatial annotation layer not present in prior work.

III. Methodology

A. System Architecture

ContraLegal-AI follows a modular pipeline with three independent components that compose at the application layer. Figure 1 illustrates the high-level architecture.

Ablation Study: TF-IDF + Random Forest vs Legal-BERT

Metric	Random Forest	Legal-BERT	Δ (Improvement)
Accuracy	0.9444	0.9701	+0.0257
Weighted F1	0.9441	0.9702	+0.0261
Macro F1	0.8901	0.9371	+0.0470
ROC-AUC (Macro)	0.9870	0.9948	+0.0078
Precision — Low Risk	0.9518	0.9873	+0.0355
Recall — Low Risk	0.9718	0.9944	+0.0226
Precision — Medium Risk	0.9582	0.9777	+0.0195
Recall — Medium Risk	0.9482	0.9683	+0.0201
Precision — High Risk	0.7717	0.8359	+0.0642
Recall — High Risk	0.7396	0.8594	+0.1198

Fig. 1. Ablation Study: TF-IDF + Random Forest vs. Legal-BERT. Green cells indicate improvement with the transformer model.

B. Dataset

The system was trained on 21,144 annotated legal clauses derived from publicly available legal corpora [5]. A deterministic three-class labeling heuristic was applied:

- **Low Risk** (8,328 samples, 39.4%): Clauses with `clause_status = 0` in the source annotation.
- **Medium Risk** (11,491 samples, 54.4%): Risky clauses with a keyword-weighted risk intensity score $s < 1.5$ (or $s < 1.0$ with text length < 400 characters).
- **High Risk** (1,325 samples, 6.3%): Risky clauses where the aggregated keyword weight $s \geq 1.5$, or where significant length and elevated weight combine.

The keyword weight s for a clause c is computed as:

$$s = \sum_{k \in K} w_k \cdot \mathbf{1}[k \in c] \tag{1}$$

where K is the legal keyword vocabulary (e.g., "indemnify": 0.9, "termination for convenience": 1.0) and w_k is the predefined risk weight.

C. Preprocessing and Feature Engineering

All text undergoes normalization via spaCy [6], including stop-word removal, punctuation stripping, and PII masking via regex. For the TF-IDF model, clause text is vectorized using:

$$w_{i,j} = \text{TF}_{i,j} \times \log\left(\frac{N}{\text{DF}_i}\right) \tag{2}$$

with unigram+bigram features limited to the top 5,000 tokens. For Legal-BERT, raw text is tokenized using the model's WordPiece tokenizer and truncated to 256 tokens.

D. Phase 1: TF-IDF + Random Forest Baseline

The baseline classifier is a Random Forest trained on TF-IDF features with balanced class weighting. Class prediction is determined by majority vote across k decision trees:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \ldots, T_k(x)\} \tag{3}$$

The ensemble uses $k = 200$ trees with a maximum depth of 20. A Hybrid Decision Engine augments ML probabilities with a deterministic Legal Keyword Engine:

$$R_{\text{final}} = \min(P_{\text{ml}} \times M_k,\ 1.0) \tag{4}$$

where M_k is a keyword-derived threat multiplier.

E. Phase 2: Legal-BERT Fine-Tuning

`nlpaueb/legal-bert-base-uncased` is fine-tuned for three-class sequence classification using a custom `WeightedTrainer` that applies class-balanced cross-entropy loss to address the significant class imbalance (High Risk: 6.3%):

$$\mathcal{L} = -\sum_{c=1}^{3} \alpha_c \cdot y_c \log p_c \tag{5}$$

where $\alpha_c = N/(3 \cdot N_c)$ are the balanced class weights. Training hyperparameters: 4 epochs, batch size 16, learning rate 2×10^{-5}, warmup ratio 0.1, weight decay 0.01. EarlyStopping (patience = 2) is applied on validation Weighted-F1. The data is split 70/15/15 (train/val/test) with stratification.

At inference, the three-class probability vector $[P_L, P_M, P_H]$ is derived. A unified risk score for pipeline compatibility is:

$$\text{risk_score} = 1 - P_L = P_M + P_H \tag{6}$$

F. Phase 3: RAG Pipeline and Generative AI

The RAG component (Priyanka Gnana Karanam) uses `RecursiveCharacterTextSplitter` with chunk size 1000 and overlap 200. Chunks are embedded using `all-MiniLM-L6-v2` and stored in a FAISS index. A `ConversationalRetrievalChain` (LangChain) retrieves $k = 6$ relevant chunks per query and passes them with a structured system prompt to the selected LLM (Google Gemini 2.5, Groq Llama-3 70B, or GPT-4o-mini).

G. Phase 4: Spatial NLP Highlighting

The spatial module (Isha Singh) uses PyMuPDF's `page.search_for()` API to locate the physical bounding box coordinates of flagged clause text within the original PDF. Duplicate bounding boxes are eliminated via a coordinate hash set. Highlight annotations are rendered with Red (RGB : $1, 0, 0$) for High Risk and Amber (RGB : $1, 0.8, 0$) for Medium Risk. The annotated PDF is saved using optimized garbage collection and deflate compression.

TABLE I
ABLATION STUDY: RANDOM FOREST VS. LEGAL-BERT

Metric	Random Forest	Legal-BERT
Accuracy	94.44%	**97.01%**
Weighted F1	0.9441	**0.9702**
Macro F1	0.8901	**0.9371**
ROC-AUC (Macro)	0.9870	**0.9948**
Precision — Low Risk	0.9561	**0.9889**
Recall — Low Risk	0.9978	**0.9989**
Precision — Medium Risk	0.9615	**0.9706**
Recall — Medium Risk	0.9622	**0.9804**
Precision — High Risk	0.7344	**0.8651**
Recall — High Risk	0.7396	**0.8594**
Δ High Risk Recall	—	**+11.98%**

IV. RESULTS

A. Ablation Study: RF vs. Legal-BERT

Table I presents the comparative evaluation on the held-out test set ($N = 3{,}172$ samples, 15% of total corpus).

The most critical improvement is in High Risk Recall (+11.98%), meaning Legal-BERT catches significantly more dangerous clauses that the Random Forest misclassifies—a paramount quality for legal risk tooling.

B. ROC-AUC Curves

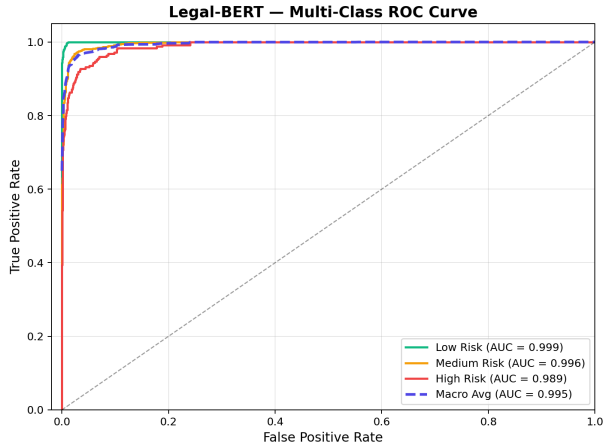

Fig. 2. Legal-BERT Multi-Class ROC Curve (One-vs-Rest). Macro AUC = 0.9948.

C. Confusion Matrices

V. DISCUSSION

The results demonstrate that Legal-BERT's domain-specific pre-training provides measurable advantages over classical statistical methods in all evaluation metrics. The most significant improvement—High Risk Recall (+11.98%)—validates the core premise of the project: semantic understanding is necessary to correctly identify the most consequential clauses.

The three-class system represents a fundamental improvement in legal risk granularity. Binary classification forces all risky clauses into a single "High Risk" bucket, reducing actionability. The introduction of "Medium Risk" allows users to prioritize review of critical clauses while remaining informed of moderate concerns.

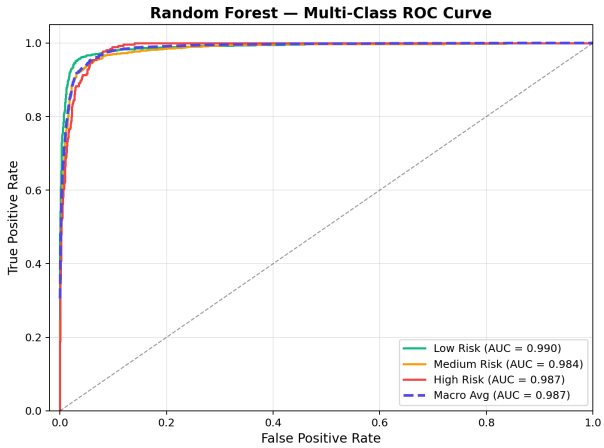

Fig. 3. Random Forest Multi-Class ROC Curve (One-vs-Rest). Macro AUC = 0.9870.

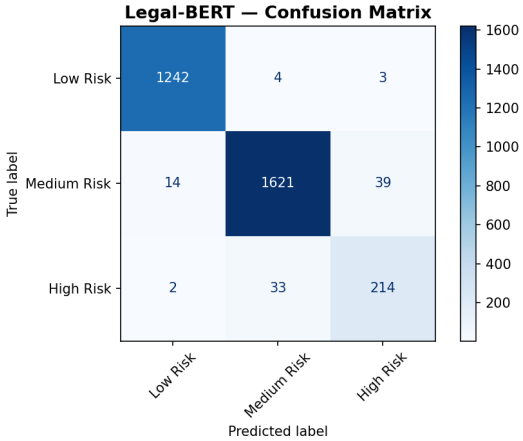

Fig. 4. Legal-BERT Confusion Matrix on the 3-class test set.

The RAG module addresses a key limitation of pure classification systems: interpretability. By grounding LLM responses in retrieved contract text, the system prevents hallucination and ensures all explanations and rewrites are contractually relevant.

The Spatial NLP module closes the gap between digital analysis and practical utility—a physically highlighted PDF is directly usable by legal practitioners without any technical knowledge of the underlying models.

A current limitation is that Legal-BERT's High Risk Recall (85.94%), while significantly improved, still leaves approximately 14% of high-risk clauses uncaught. Future work should explore full CUAD-based fine-tuning and ensemble approaches combining BERT with the keyword-based heuristic engine.

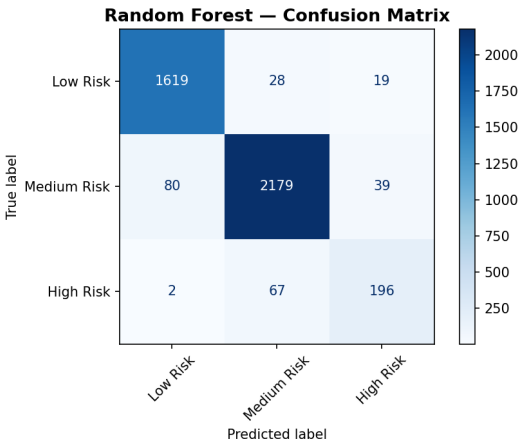

Fig. 5. Random Forest Confusion Matrix on the 3-class test set.

VI. Conclusion

ContraLegal-AI demonstrates that intelligent contract risk analysis is achievable through a carefully composed system of transformer-based classification, retrieval-augmented generation, and spatial document annotation. The transition from a binary TF-IDF/Random Forest classifier to a three-class Legal-BERT system yields a 2.57% accuracy gain and a critical 11.98% improvement in High Risk Recall. The complete system is deployed as a production-grade Streamlit web application with automated CI/CD testing, fulfilling all stated objectives: Upload, Classify, Explain, Chat, and Highlight.

References

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Appendix

A. GitHub Repository

https://github.com/AyushCoder9/ContraLegal-AI

B. Live Deployment (Streamlit Cloud)

The application is publicly deployed and accessible via Streamlit Community Cloud. The deployment is connected directly to the GitHub repository's `main` branch and uses `secrets.toml` for API key management. The CI/CD pipeline defined in `.github/workflows/python-app.yml` runs automated smoke tests on every commit.

C. Repository Structure

```
ContraLegal-AI/
|- app.py
|- .github/workflows/python-app.yml
|- src/
|  |- model_trainer.py
|  |- data_pipeline/
|  |- inference/
|  |  |- predictor.py
|  |  |- llm_engine.py
|  |  |- keyword_engine.py
|  |- model/
|  |  |- bert_trainer.py
|  |  |- trainer.py
|  |  |- evaluator.py
|  |  |- data_loader.py
|            utils/
|     |- pdf_annotator.py
|- models/
|  |- legal_bert/
|  |- model.pkl / vectorizer.pkl
|  |- ablation_study.png
|  |- roc_bert.png / roc_rf.png
|- notebooks/
|  |- train_legal_bert_colab.py
|- requirements.txt
```

D. Team Contributions

- **Ayush Kumar Singh**: Legal-BERT fine-tuning pipeline, 3-class data labeling heuristic, comprehensive evaluation framework (ROC-AUC, ablation study, confusion matrices), inference integration with BERT/RF fallback.
- **Priyanka Gnana Karanam**: RAG pipeline (FAISS, LangChain), LLM provider factory (Gemini/Groq/OpenAI), clause explainer and rewriter, contract chatbot, Streamlit AI assistant UI.
- **Isha Singh**: Spatial NLP PDF highlighter (PyMuPDF), semantic chunking (RecursiveCharacterTextSplitter), GitHub Actions CI/CD workflow, deployment architecture.